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Semiconductor apparatus and heating device in semiconductor apparatus

Abstract

The present disclosure discloses a heating device in a semiconductor apparatus and the semiconductor apparatus, including a heating body configured to carry a wafer, a heating member configured to generate heat being arranged in the heating body; and a cooling structure, which is arranged on the heating body below, and a cooling structure being arranged below the heating body. The cooling structure is configured to perform heat exchange with the heating body selectively at different positions away from the heating body. The heating device in the semiconductor apparatus and the semiconductor apparatus of the present disclosure are configured to expand an application temperature range of the heating device to satisfy different temperature requirements.

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Background/Summary**CROSS-REFERENCES TO RELATED APPLICATIONS**

(1) This application is a national phase entry under 35 U.S.C. § 371 of International Application No. PCT/CN2021/084863, filed on Apr. 1, 2021, which claims priority to Chinese Application No. 202010281032.9 filed on Apr. 10, 2020, the entire content of all of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure generally relates to the field of semiconductor apparatus and, more particularly, to a heating device in a semiconductor apparatus and a semiconductor apparatus.

BACKGROUND

(3) With the continuous development of the semiconductor industry, semiconductor processing is more diverse. However, no matter what kind of process, temperature is an important part of it. Temperature of the process directly affects the apparatus capacity and cost. Generally, a wafer is placed on a heater of a process chamber for processing. During this process, the temperature of the heater is required to remain unchanged. Currently, a method of simultaneously controlling heating and cooling is usually used to keep the temperature unchanged.

(4) However, in a physical vapor deposition (PVD) process, the temperature is difficult to be kept unchanged, because the PVD process is to deposit a sputtered target material onto the wafer by a sputtering deposition technology. Since the temperature of the sputtered target material is high, the target material may transfer its own heat through the wafer to the heater that carries the wafer during deposition. Thus, the surface temperature of the heater increases. As process time increases, the surface temperature of the heater gets higher and higher. Although the heater can be cooled by a cooling device arranged under the heater, the following problems may be caused.

(5) Since heat exchange efficiency of the cooling device is fixed, if heat dissipated by the cooling device is more than the heat generated by the heater, the temperature of the heater cannot be maintained at a relatively high temperature. If the heat dissipated by the cooling device is less than the heat increased by the heater, the temperature of the heater cannot be maintained at a relatively low temperature. Thus, the cooling device may limit a temperature application range of the heater, which cannot satisfy temperature requirements.

SUMMARY

(6) The purpose of the present disclosure is to provide a heating device in a semiconductor apparatus and the semiconductor apparatus to expand the application temperature range of the heating device to satisfy different temperature requirements.

(7) To achieve the above object, embodiments of the present disclosure provide a heating apparatus in a semiconductor apparatus, which is arranged in a process chamber of the semiconductor apparatus and includes:

(8) a heating body configured to carry a wafer, a heating member configured to generate heat being

arranged in the heating body; and a cooling structure arranged below the heating body, the cooling structure being configured to selectively perform heat exchange with the heating body at different positions away from the heating body.

(9) In some embodiments, the cooling structure includes: a cooling body, a cooling member configured to perform heat exchange with the heating body being arranged in the cooling body; and a cooling ascending and descending assembly connected to the cooling body and configured to drive the cooling body to ascend and descend to adjust a distance between the cooling body and the heating body in a vertical direction.

(10) In some embodiments, the heating device further includes a connection member and a sealing structure, wherein: the connection member is connected to the heating body, the connection member and the heating body form accommodation space at a bottom of the heating body, the cooling body is located in the accommodation space, and the cooling ascending and descending assembly is extended to outside of the process chamber passing through a through-hole arranged at a bottom of the process chamber; and an upper end of the sealing structure is sealed with and connected to the connection member, and a lower end of the sealing structure is sealed with and connected to the bottom of the process chamber to seal the accommodation space and the through-hole.

(11) In some embodiments, the sealing structure includes a bellows and a bellows shaft, wherein: an upper end of the bellows shaft is connected to the connection member, and a lower end of the bellows shaft is extended to the outside of the process chamber passing through the through-hole to be connected to an ascending and descending drive source; and the bellows is sleeved at the bellows shaft, an upper end of the bellows is sealed with and connected to the connection member, and a lower end of the bellows is sealed with and connected to the bottom of the process chamber to seal the accommodation space and the through-hole.

(12) In some embodiments, the cooling ascending and descending assembly includes a lift member, at least one screw, and at least one elastic member, wherein hollow space communicating with the accommodation space is arranged in the bellows shaft, the lift member is located in the hollow space, and an upper end of the lift member is connected to the cooling body; at least one threaded hole is arranged at a bottom surface of the lift member, at least one mounting hole is arranged at an inner surface of the bellows shaft opposite to the bottom surface of the lift member, and the mounting hole is coaxially arranged with the threaded hole in a one-to-one correspondence; the screw is threadedly connected to the corresponding threaded hole passing through the mounting hole from the bottom of the bellows shaft to a top in a one-to-one correspondence; and the elastic member is arranged in a one-to-one correspondence with the screw, is located between the bottom surface of the lift member and the inner surface of the bellows shaft, and is in a compressed state.

(13) In some embodiments, lead-out space and two first lead-out through-holes communicating with the lead-out space are arranged in the lift member, two second lead-out through-holes communicating with the hollow space are arranged in the bellows shaft, and the two second lead-out through-holes are coaxially arranged with the two first lead-out through-holes; and the cooling member further includes a cooling pipeline configured to transfer cooling water, a water inlet pipeline, and a water outlet pipeline, an end of the water inlet pipeline and an end of the water outlet pipeline are connected to an inlet and an outlet of the cooling pipeline, respectively, and the other end of the water inlet pipeline and the other end of the water outlet pipeline are extended the outside of the process chamber passing through the corresponding first lead-out through-hole and the second lead-out through-hole in sequence.

(14) In some embodiments, the cooling structure includes a plurality of cooling bodies, the plurality of cooling bodies being arranged at intervals in a direction away from a bottom surface of the heating body, and a cooling member configured to perform heat exchange with the heating body being arranged in each of the cooling bodies.

(15) In some embodiments, the heating device further includes a connection member and a sealing

structure, wherein: the connection member is connected to the heating body, the connection member forms accommodation space with the heating body at a bottom of the heating body, and the plurality of cooling bodies are located in the accommodation space, wherein each of the cooling members includes a cooling pipeline configured to transfer cooling water, a water inlet pipeline, and a water outlet pipeline, an end of the water inlet pipeline and an end of the water outlet pipeline are connected to an inlet and an outlet of the cooling pipeline, respectively, and the other end of the water inlet pipeline and the other end of the water outlet pipeline are extended to the outside of the process chamber passing through the through-hole arranged at the bottom of the process chamber; and an upper end of the sealing structure is sealed with and connected to the connection member, and a lower end of the sealing structure is sealed with and connected to the bottom of the process chamber to seal the accommodation space and the through-hole.

(16) In some embodiments, the sealing structure includes a bellows and a bellows shaft, wherein: an upper end of the bellows is connected to the connection member, a lower end of the bellows shaft is extended to the outside of the process chamber passing through the through-hole to be connected to an ascending and descending drive source, the bellows shaft is provided with hollow space communicating with the accommodation space and a plurality of pairs of lead-out through-holes communicating with the hollow space, and the other end of the water inlet pipeline and the other end of the water outlet pipeline are extended to the outside of the process chamber through a pair of corresponding lead-out through-holes; and the bellows is sleeved at the bellows shaft, an upper end of the bellows is sealed with and connected to the connection member, and a lower end of the bellows is sealed with and connected to the bottom of the process chamber to seal the accommodation space and the through-hole.

(17) In some embodiments, an upper surface of a most upper cooling body is attached to an upper surface of the heating body.

(18) In some embodiments, a lower surface of the heating body is parallel to an upper surface of the cooling body.

(19) In some embodiments, the heating device further includes a heat exchange gas inlet pipeline, a gas outlet end of the heat exchange gas inlet pipeline is communicated with the accommodation space, and a gas inlet end of the heat exchange gas inlet pipeline is configured to be connected to a heat exchange gas source.

(20) As another technical solution, embodiments of the present disclosure further provide a semiconductor apparatus, including a process chamber, and further including the heating device of embodiments of the present disclosure, and the heating device is arranged in the process chamber.

(21) The beneficial effects of the present disclosure include as follows.

(22) The heating device in the semiconductor apparatus provided by embodiments of the present disclosure may perform heat exchange with the heating body selectively at different positions away from the heating body through the cooling structure arranged below the heating body. Thus, the cooling structure may cool the heating body with different heat exchange efficiencies. Thus, the heat exchange may be performed at a position close to the heating body to control the temperature of the heating body in a relatively low-temperature range, and heat exchange may be performed at a position far away from the heating body to control the temperature of the heating body in a relatively high-temperature range to further expand the application temperature range of the heating device to satisfy different temperature requirements.

(23) In the semiconductor apparatus provided by embodiments of the present disclosure, by using the above-mentioned heating device provided by embodiments of the present disclosure, the application temperature range of the heating device may be expanded to satisfy different temperature requirements.

(24) The present disclosure may have other features and advantages. These features and advantages are obvious from the accompanying drawings combined in the specification and the subsequent specific embodiments or are described in detail in the accompanying drawings combined in the

specification and the subsequent specific embodiments. These accompanying drawings and specific embodiments are used to explain the specific principle of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other objects, features, and advantages of the present disclosure will become more apparent by performing a more detailed description of exemplary embodiments of the present disclosure in connection with the accompanying drawings. In exemplary embodiments of the present disclosure, same reference numerals generally refer to the same components.

(2) FIG. 1 is a schematic structural diagram of a heating device according to some embodiments of the present disclosure.

(3) FIG. 2 is a schematic structural diagram of a heating device of a semiconductor apparatus according to Embodiment 1 of the present disclosure.

(4) FIG. 3 is a schematic structural diagram of a heating device of a semiconductor apparatus according to Embodiment 2 of the present disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(5) FIG. 1 shows a heating device, which is configured to solve a problem that temperature on a surface of the heating device continuously increases. As shown in FIG. 1, the heating device is arranged in a process chamber (not shown in the figure) and includes a heating body **101**, a cooling body **103**, a heating coil **102**, a cooling water channel **104**, a water inlet pipeline **105a**, a water outlet pipeline **105b**, a connection cylinder **107**, and a fixing base **106**. The heating coil **102** may be arranged in the heating body **101** and configured to generate heat. The cooling water channel **104** may be arranged in the cooling body **103** and configured to exchange heat with the cooling body **103** by transmitting cooling water to indirectly cool the heating body **101**. An upper end of the connection cylinder **107** may be sealed with and connected to the cooling body **103**. A lower end of the connection cylinder **107** may be sealed with and connected to the fixing base **106**. The fixing base **106** may be configured to be fixedly connected to a bottom of the process chamber.

(6) An inlet and an outlet of the cooling water channel **104** may be connected to one end of the water inlet pipeline **105a** and one end of the water outlet pipeline **105b**, respectively. and the other ends of the water inlet pipeline **105a** and the water outlet pipeline **105b** may extend outside of the process chamber through internal space **108** of the connection cylinder **107**, a through-hole **106a** in the fixing base **106**, and a through-hole at the bottom of the process chamber (not shown in the figure) in sequence to be connected to a cooling water source. In addition, the heating device may further include a temperature measurement device **100**. A detection end of the temperature measurement device **100** may be in contact with the heating body **101**. A lead wire end may be led out to the outside of the process chamber through the same path as the above-mentioned water inlet pipeline **105a** and water outlet pipeline **105b**.

(7) When the above heating device is used, heat generated by the heating coil **102** may be used to maintain a process temperature of the heating body **101**. By introducing cooling water into the cooling water channel **104**, temperature rise of the heating body **101** caused by the processing process may be lowered. In this process, when the temperature of the heating body **101** is not higher than the process temperature, the heating coil **102** may work. When the temperature of the heating body **101** is higher than the process temperature, cooling water may be introduced into the cooling water channel **104** to maintain the temperature of the heating body **101** at the process temperature.

(8) However, in practical applications, it is found that the above-mentioned heating device can only maintain the temperature of the heating body **101** in a relatively low-temperature range (for example, less than 100° C.). When the required process temperature is higher than this temperature

range, since the thermal conductivity of the cooling body **103** is relatively high, after the cooling water is introduced into the cooling water channel **105**, the heat of the heating body **101** may be dissipated away quickly. If a speed of dissipating the heat is greater than a speed of generating heat by the heating coil **102** at a full power output, the temperature of the heating body **101** cannot reach a relatively high process temperature. Thus, A high-temperature requirement cannot be satisfied.

(9) Another heating device may be similar to the structure of the above-mentioned heating device shown in FIG. **1**. A difference between the two is that a certain gap may be arranged between the upper surface of the cooling body and the lower surface of the heating body, which reduces heat exchange efficiency between the cooling body and the heating body. Thus, the heat of the heating body may not be dissipated greatly to cause the temperature of the heating body to reach a relatively high process temperature. However, in practical applications, it is found that the heating device can only maintain the temperature of the heating body at a relatively high-temperature range (e.g., greater than 80° C.). When the required process temperature is lower than this temperature range, since the heat exchange efficiency of the cooling body is relatively low, a speed of dissipating the heat may be lower than a speed of increasing the heat of the heating body.

Therefore, the temperature of the heating body cannot be controlled at a lower process temperature, and the low-temperature requirement cannot be met. In addition, when the apparatus needs to be maintained, the temperature of the heating body may need to be lowered to a normal temperature state to facilitate personnel to perform maintenance. However, since the heat exchange efficiency of the cooling body is low, the time to reduce the heating body to the normal temperature may be too long. Thus, the apparatus maintenance time may be too long, which may affect the apparatus utilization rate.

(10) In order to solve the above problems, the present disclosure provides a heating device in a semiconductor apparatus. The heating device may be arranged in the process chamber of the semiconductor apparatus and include a heating body and a cooling structure. The heating body may be configured to carry a wafer. The heating body that is configured to generate heat may be configured in the heating body. The cooling structure may be arranged below the heating body. The cooling structure may be configured to selectively perform heat exchange with the heating body at different positions from the heating body.

(11) The cooling structure may perform the heat exchange with the heating body at different positions from the heating body in a plurality of manners. For example, an overall height of the above-mentioned cooling structure may be adjusted to change the overall position of the cooling structure from the heating body. For another example, the overall height of the above cooling structure may remain unchanged, and the heat exchange efficiency may be changed only by selecting the cooling components (the components that perform heat exchange with the heating body) at different positions in the cooling structure to work.

(12) By selectively exchanging heat with the heating body at different positions away from the heating body by the cooling structure, the cooling structure may cool the heating body with different heat exchange efficiencies. Thus, heat may be exchanged with the heating body at a position close to the heating body to control the temperature of the heating body in a low-temperature range, and heat may be exchanged with the heating body at a position far away from the heating body to control the temperature of the heating body in a high-temperature range. Thus, an application temperature range of the heating device may be expanded, which satisfies the high-temperature requirement and also the low-temperature requirement.

(13) The present disclosure may be further described in detail below with reference to the accompanying drawings and specific embodiments.

Embodiment 1

(14) With reference to FIG. **2**, Embodiment 1 of the present disclosure provides a heating device in a semiconductor apparatus. The heating device includes a heating body **301** and a cooling structure. The heating body **301** may be configured to carry a wafer **110**. For example, the wafer **110** may be

placed on an upper surface of the heating body **301**. The heating body **301** may be in a circular plate shape. Of course, another arbitrary shape may also be used according to specific needs. Moreover, a heating member **302** configured to generate heat may be arranged in the heating body **301**. The heating member **302** may include a heating element capable of generating heat, such as a heating resistance wire, a heating lamp, etc. Specifically, the heating member **302** shown in FIG. 2 is a heating resistance wire, which is embedded in the heating body **301** from the lower surface of the heating body **301** and evenly distributed relative to the lower surface of the heating body **301**. Thus, the wafer **110** may be uniformly heated.

(15) The cooling structure may be arranged below the heating body **301**. The cooling structure may include a cooling body **313** and a cooling lift assembly. A cooling member **304** configured to perform heat exchange with the heating body **301** may be arranged in the cooling body **313**. The cooling lift assembly may be connected to the cooling body **313** and configured to drive the cooling body **313** to ascend and descend to adjust a distance between the cooling body **313** and the heating body **301** in a vertical direction. That is, driven by the cooling lift assembly, an overall height of the cooling body **313** may change to adjust the position of the cooling member **304** of the cooling body **313** away from the heating body **301**. The closer the cooling member **304** to the heating body **301** is, the higher the heat exchange efficiency with the heating body **301** is, and the easier the temperature of the heating body **301** may be maintained at a low temperature. Thus, the low-temperature requirement may be satisfied. Otherwise, the further the cooling member **304** away from the heating body **301** is, the higher the heat exchange efficiency of the heating body **301** is, and the easier the temperature of the heating body **301** may be maintained at a high temperature, which satisfies the high-temperature requirement. Thus, the application temperature range of the heating device may be expanded, which satisfies different temperature needs. In addition, with the above-mentioned cooling lift assembly, the overall height of the cooling body **313** may be adjusted freely. Thus, the flexibility of temperature adjustment may be improved.

(16) In some embodiments, a lower end of the cooling lift assembly may need to be extended to the outside of the process chamber. In this case, in order to ensure the tightness of the process chamber. In some embodiments, the heating device may further include a connection member **307** and a seal structure. The connection member **307** may be connected to the heating body **301**. The connection member **307** may form accommodation space **310** with the heating body **301** at the bottom of the heating body **301**. The cooling body **313** may be located in the accommodation space **310**. The cooling lift assembly may be extended to the outside of the process chamber **400** passing through the through-hole arranged at the bottom (not shown) of the process chamber **400**. The upper end of the seal structure may be sealed and connected to the bottom of the process chamber **400**. The seal structure may be configured to seal the accommodation space **310** and the through-hole. An internal part of a dashed line block may represent an internal part of the process chamber **400**. An outer part of the dashed line block may represent outside of the process chamber.

(17) In some embodiments, in order to facilitate disassembly and assembly, and to protect the heating body **301** from being damaged during disassembly and assembly, the connection member **307** may be fixed at the bottom of the heating body **301** through an annular connection piece **306**. The annular connection piece **306** may be sealed with and connected to the heating body **301** and the connection member **307**, respectively. In this case, the above-mentioned annular connection piece **306**, the connection member **307**, and the heating body **301** may form the above-mentioned accommodation space **310**. Various sealing and connection manners may be included, for example, welding, or threaded connection with vacuum sealing treatment. Of course, in practical applications, the connection member **307** may also be directly connected to the heating body **301**.

(18) The above sealing structure may include various structures. For example, as shown in FIG. 2, in this embodiment, the sealing structure includes a bellows **308** and a bellows shaft **305**. An upper end of the bellows shaft **305** may be connected to the connection member **307**, and a lower end of the bellows shaft **305** may extend to the outside of the process chamber **400** by passing through the

through-hole and may be configured to be connected to a lift drive source (not shown in the figure). Driven by the lift drive source, the bellows shaft **305** may be configured to drive the heating body **301** to ascend and descend through the connection member **307**. Thus, the heating body **301**, the connection member **307**, and the cooling body **313** may all ascend and descend synchronously. The bellows **308** may be sleeved on the bellows shaft **305**. The upper end of the bellows **308** may be sealed with and connected to the connection member **307**. The lower end of the bellows **308** may be sealed with and connected to the bottom of the process chamber **400** and configured to seal the accommodation space **310** and the above-mentioned through-hole. Thus, the sealing performance of the process chamber **400** may be ensured.

(19) In some embodiments, the lower end of the above-mentioned bellows **308** may be sealed with and connected to a bottom wall (not shown in the figure) of the process chamber **400** through a fixing base **309**. A fixing base through-hole **309** may be arranged in the fixing base **309**. The lower end of the bellows shaft **305** may extend to the outside of the process chamber **400** through the fixed base through-hole **309a** and the through-hole at the bottom of the process chamber **400** in sequence. It can be understood that the above-mentioned bellows **308** may be also configured to seal the fixing base through-hole **309a**.

(20) The above-mentioned cooling lift assembly may include various structures. For example, in this embodiment, the cooling lift assembly may include a lift member **303**, at least one screw **401**, and at least one elastic member **402**. Hollow space **305a** that is communicated with the accommodation space **310** may be arranged in the bellows shaft **305**. The lift member **303** may be located in the hollow space **305a**. An upper end of the lift member **303** may be connected to the above cooling body **313**. A connection manner of the lift member **303** and the cooling body **313** may include, for example, an integration manner, a welding manner, etc. At least one threaded hole **403** may be arranged at a bottom surface of the lift member **303**. At least mounting hole **404** may be arranged on an inner surface of the bellows shaft **305** opposite to the bottom surface of the lift member **303**. The mounting holes **404** may be coaxially arranged with the threaded holes **403** in a one-to-one correspondence. The screws **401** may be threadedly connected to the threaded holes **403** passing through the mounting holes **404** in a one-to-one correspondence from bottom to top. The elastic members **402** may be arranged in a one-to-one correspondence with the screws **401**, may be located between the bottom surface of the lift member **303** and the inner surface of the bellows shaft **305**, and may be in a compressed state.

(21) In some embodiments, a plurality of screws **401** may be included (two screws **401** are shown in FIG. 2). The plurality of screws **401** may be arranged symmetrically relative to an axis of the bellows shaft **305**. By arranging the plurality of screws **401**, the stability of supporting the cooling body **313** may be improved.

(22) When the above-mentioned screw **401** is rotated, since the screw is threadedly connected to the threaded hole **403** at the lift member **303**. With the thread cooperation, the screw **401** may be fixed. The lift member **303** may ascend or descend along a axial direction of the screw **401** to drive the cooling body **313** to ascend or descend synchronously. Meanwhile, since the elastic member **402** is in the compressed state, the elastic member **402** may always apply an upward elastic force to the lift member **303** to support the lift member **303** and the cooling body **313**. The elastic member **402** may include, for example, a compression spring.

(23) It should be noted that the cooling lift assembly may not be limited to the above structure of this embodiment. In practical applications, the cooling lift assembly may further adopt any other structure that can drive the cooling body **313** to ascend and descend.

(24) It should also be noted that, in this embodiment, the heating body **301** may realize ascending and descending using a heating ascending and descending assembly. However, embodiments of the present disclosure may be not limited to this. In practical applications, according to specific needs, the above heating ascending and descending assembly may not be provided. That is, the heating body **301** may be fixed and may not move relative to the process chamber. In this case, if the

cooling lift assembly needs to be extended out of the process chamber, a corresponding sealing structure may be used to seal the through-hole at the bottom of the process chamber, which is configured for the cooling lift assembly to pass through to ensure the sealing performance of the process chamber. Of course, the above cooling and lifting assemblies may also be arranged inside the process chamber. In this case, an automatic control method may be used to provide ascending and descending power for the cooling and lifting assemblies.

(25) In this embodiment, lead-out space **303a** and two first lead-out through-holes communicating with the lead-out space **303a** may also be arranged in the lifting member **303**. Moreover, two second lead-out through-holes communicating with the hollow space **305a** may be arranged in the bellows shaft **305**. The two second lead-out through-holes may be coaxially arranged with the two first lead-out through holes. The above-mentioned cooling member **304** may include a cooling pipeline **304a** configured to transfer cooling water, a water inlet pipeline **304b**, and a water outlet pipeline **304c**. One end of the water inlet pipeline **304b** and one end of the water outlet pipeline **304c** may be connected to the inlet and the outlet of the cooling pipeline **304a**, respectively. The other end of the water inlet pipeline **304b** and the other end of the water outlet pipeline **304c** may be extended to the outside of the process chamber **400** through the corresponding first lead-out through-holes and the second lead-out through-holes in sequence to be able to provide a water source connection of the cooling water.

(26) In some embodiments, the above-mentioned cooling pipeline **304a** may be an annular water pipe or may also be a helical water pipe, which is evenly distributed relative to the upper surface of the cooling body **313** to improve cooling uniformity.

(27) In this embodiment, In some embodiments, the lower surface of the heating body **301** and the upper surface of the cooling body **313** may be parallel to each other. Thus, heat exchange efficiencies between the upper surface of the cooling body **313** at different positions and the heating body **301** can be the same. Thus, the temperature uniformity of the heating body **301** may be improved.

(28) In this embodiment, In some embodiments, the heating device may further include a heat exchange gas input pipeline (not shown in the figure). A gas outlet end of the heat exchange gas input pipeline may be communicated with the above-mentioned accommodation space **310**. An inlet end of the heat exchange gas input pipeline may be connected to a heat exchange gas source. By introducing the heat exchange gas to the accommodation space **310**, a gas pressure in the accommodation space **310** may be improved, which may further improve the heat exchange efficiency and improves the cooling efficiency. Especially, when the apparatus needs maintenance, time for reducing the heating body to a normal temperature may be shortened, which reduces apparatus maintenance time to improve the apparatus utilization rate. The heat exchange gas may be, for example, compression gas (e.g., high-pressure air).

Embodiment 2

(29) Referring to FIG. 3, Embodiment 2 of the present disclosure provides a heating device in the semiconductor apparatus. Compared with Embodiment 1 above, the heating device also includes a heating body **301** and a cooling structure. The difference may include that the specific structure of the cooling structure is different.

(30) Specifically, in this embodiment, the cooling structure may include a plurality of cooling bodies. The plurality of cooling bodies may be arranged at intervals along a direction away from the lower surface of the heating body **301**. A cooling member configured to perform heat exchange with the heating body **301** may be arranged in each cooling body. Since different cooling members have different positions from the heating body **301**, by selectively controlling the cooling member of the at least one cooling body to work, cooling may be performed on the heating body **301** with different heat exchange efficiencies to satisfy different temperature needs. That is, positions of the cooling bodies from the heating body **301** may be unchanged. The heat exchange efficiency may be changed only by selecting the cooling members at the different positions to work.

(31) In some embodiments, the upper surface of the uppermost cooling body may be attached to the upper surface of the heating body **301**. Thus, the heat exchange efficiency between the uppermost cooling body and the heating body **301** may be further improved to satisfy the high cooling speed requirement.

(32) For example, FIG. **3** shows two cooling bodies, which are a first cooling body **303A** and a second cooling body **303B** located below the first cooling body **303A**. The second cooling body **303B** may be arranged with the first cooling body **303A** at an interval. A first cooling member **304A** may be arranged in the first cooling body **303A**. A second cooling member **304B** may be arranged in the second cooling body **303B**.

(33) The first cooling member **304A** and the second cooling member **304B** may operate individually or simultaneously. When the first cooling member **304A** and the second cooling member **304B** work simultaneously, the heat exchange efficiency may be the highest. Since the first cooling member **304A** is closer to the second cooling member **304B** because the first cooling part **304A** is closer to the heating body **301** than the second cooling member **304B**. The heat exchange efficiency of the first cooling member when the first cooling member **304A** operates individually may be higher than the heat exchange efficiency of the second cooling member **304B** when the second cooling member **304B** operates individually. Therefore, by selecting at least one of the first cooling member **304A** and the second cooling member **304B**, switching may be performed among the three types of the heat exchange efficiencies to satisfy different temperature requirements. It is easy to understand that the greater the number of cooling bodies is, the more choices of the heat exchange efficiency have, and the greater the flexibility of temperature regulation is.

(34) In this embodiment, the heating device may also include a heating ascending and descending assembly configured to drive the heating body **301** to ascend and descend. The heating ascending and descending assembly may adopt a structure similar to the above-mentioned Embodiment 1 (refer to the relevant part of FIG. **2**). A difference may include that, as shown in FIG. **3**, in this embodiment, an upper end of a connection member **307'** is sealed with and connected to the first cooling body **303A** and form accommodation space **310'** at the bottom of the first cooling body **303A** with the first cooling body **303A**. The second cooling body **303B** may be located in the accommodation space **310'** and may be integrated with the connection member **307'**. That is, the above-mentioned connection member **307'** may be used as the second cooling body **303B**. The second cooling member **304B** may be arranged in the connection member **307'** to seal the process chamber.

(35) The other structures of the heating ascending and descending assembly may be the same as the above-mentioned Embodiment 1. Detailed description may be in the above-mentioned Embodiment 1 and may not be repeated here.

(36) In this embodiment, each cooling member may include a cooling pipeline configured to transfer cooling water, a water inlet pipeline, and a water outlet pipeline. The water inlet pipeline and the water outlet pipeline may be extended to the outside of the process chamber in the same manner as in the above-mentioned Embodiment 1. Taking the first cooling member **304A** and the second cooling member **304B** shown in FIG. **3** as an example, the water inlet pipeline **A1** and the water outlet pipeline **A2** in the first cooling part **304A**, and the water inlet pipeline **B1** and the water outlet pipeline **B2** in the second cooling member **304B** may all be extended to the outside of the process chamber through the hollow space of the bellows shaft (not shown in the figure).

(37) Other structures of the heating device of Embodiment 2 of the present disclosure are similar to those in Embodiment 1 above. The detailed description is included in the above-mentioned Embodiment 1 and may not be repeated here.

(38) In summary, In the heating device in the semiconductor apparatus provided by embodiments of the present disclosure, through the cooling structure arranged below the heating body, heat exchange may be performed selectively at different positions from the heating body. Thus, the

cooling structure may cool the heating body with different heat exchange efficiencies. Therefore, heat exchange may be performed at a position close to the heating body to control the temperature of the heating body to be in a low-temperature range. The heat exchange may be performed at a position far away from the heating body to control the temperature of the heating body to be within the high-temperature range. The application temperature range of the heating device may be expanded to satisfy different temperature requirements.

(39) Another embodiment of the present disclosure further provides a semiconductor apparatus, such as a physical vapor deposition apparatus, including a process chamber and a heating device arranged in the process chamber. The heating device may adopt the heating device provided by the above-mentioned embodiments of the present disclosure.

(40) In the semiconductor apparatus provided by embodiments of the present disclosure, by using the above-mentioned heating device provided by embodiments of the present disclosure, the application temperature range of the heating device may be expanded to satisfy different temperature requirements.

(41) Various embodiments of the present disclosure have been described above. The above descriptions are exemplary, not exhaustive, and not limited to the disclosed embodiments. Modifications and variations made without departing from the scope or spirit of scopes of embodiments of the present disclosure may be apparent to those of ordinary skill in the art.

Claims

1. A heating device in a semiconductor apparatus arranged in a process chamber of the semiconductor apparatus comprising: a heating body configured to carry a wafer, a heating member configured to generate heat being arranged in the heating body; and a cooling structure arranged below the heating body and configured to selectively perform heat exchange with the heating body at different positions away from the heating body, and including: at least one cooling body, a cooling member configured to perform heat exchange with the heating body being arranged in the at least one cooling body; and a cooling ascending and descending assembly connected to the at least one cooling body and configured to drive the at least one cooling body to ascend and descend to adjust a distance between the at least one cooling body and the heating body in a vertical direction; a connection member connected to the heating body, the connection member and the heating body forming accommodation space at a bottom of the heating body, the at least one cooling body being located in the accommodation space, and the cooling ascending and descending assembly being extended to outside of the process chamber passing through a through-hole arranged at a bottom of the process chamber; and a sealing structure, an upper end of the sealing structure being sealed with and connected to the connection member, and a lower end of the sealing structure being sealed with and connected to the bottom of the process chamber to seal the accommodation space and the through-hole.

2. The heating device according to claim 1, wherein the sealing structure includes: a bellows shaft, an upper end of the bellows shaft being connected to the connection member, and a lower end of the bellows shaft being extended to the outside of the process chamber passing through the through-hole to be connected to an ascending and descending drive source; and a bellows sleeved at the bellows shaft, an upper end of the bellows being sealed with and connected to the connection member, and a lower end of the bellows being sealed with and connected to the bottom of the process chamber to seal the accommodation space and the through-hole.

3. The heating device according to claim 2, wherein the cooling ascending and descending assembly further includes: a lift member located in hollow space, the hollow space communicating with the accommodation space being arranged in the bellows shaft, and an upper end of the lift member being connected to the at least one cooling body; a screw, a threaded hole being arranged at a bottom surface of the lift member, at least one mounting hole being arranged at an inner

surface of the bellows shaft opposite to the bottom surface of the lift member, and the mounting hole being coaxially arranged with the threaded hole in a one-to-one correspondence, the screw being threadedly connected to the corresponding threaded hole passing through the mounting hole from the bottom of the bellows shaft to top in a one-to-one correspondence; and an elastic member arranged in a one-to-one correspondence with the screw, located between the bottom surface of the lift member and the inner surface of the bellows shaft, and being in a compressed state.

4. The heating device according to claim 3, wherein: lead-out space and two first lead-out through-holes communicating with the lead-out space are arranged in the lift member; two second lead-out through-holes communicating with the accommodation space are arranged in the bellows shaft, and the two second lead-out through-holes are coaxially arranged with the two first lead-out through-holes; the cooling member further includes: a cooling pipeline configured to transfer cooling water; a water inlet pipeline; and a water outlet pipeline, an end of the water inlet pipeline and an end of the water outlet pipeline being connected to an inlet and an outlet of the cooling pipeline, respectively, and the other end of the water inlet pipeline and the other end of the water outlet pipeline being extended to the outside of the process chamber passing through the corresponding first lead-out through-hole and the second lead-out through-hole in sequence.

5. The heating device according to claim 1, wherein the at least one cooling body includes: a plurality of cooling bodies arranged at intervals in a direction away from a bottom surface of the heating body; and the cooling member configured to perform heat exchange with the heating body and being arranged in each cooling body of the plurality of cooling bodies.

6. The heating device according to claim 5, wherein: the plurality of cooling bodies is located in the accommodation space, the cooling member including; a cooling pipeline configured to transfer cooling water, a water inlet pipeline; and a water outlet pipeline, an end of the water inlet pipeline and an end of the water outlet pipeline are connected to an inlet and an outlet of the cooling pipeline, respectively, and the other end of the water inlet pipeline and the other end of the water outlet pipeline are extended to the outside of the process chamber passing through a through-hole arranged at the bottom of the process chamber.

7. The heating device according to claim 6, wherein the sealing structure includes: a bellows shaft, an upper end of the bellows shaft being connected to the connection member, a lower end of the bellows shaft being extended to the outside of the process chamber passing through the through-hole to be connected to an ascending and descending drive source, the bellows shaft being provided with hollow space communicating with the accommodation space and a plurality of pairs of lead-out through-holes communicating with the accommodation space, and the other end of the water inlet pipeline and the other end of the water outlet pipeline being extended to the outside of the process chamber through a pair of corresponding lead-out through-holes; and a bellows sleeved at the bellows shaft, an upper end of the bellows being sealed with and connected to the connection member, and a lower end of the bellows being sealed with and connected to the bottom of the process chamber to seal the accommodation space and the through-hole.

8. The heating device according to claim 5, wherein an upper surface of a most upper cooling body of the plurality of cooling bodies is attached to a lower surface of the heating body.

9. The heating device according to claim 5, wherein a lower surface of the heating body is parallel to an upper surface of the plurality of cooling bodies.

10. The heating device according to claim 1, further comprising: a heat exchange gas inlet pipeline, a gas outlet end of the heat exchange gas inlet pipeline being communicated with the accommodation space, and a gas inlet end of the heat exchange gas inlet pipeline being configured to be connected to a heat exchange gas source.

11. A semiconductor apparatus comprising: a process chamber; and a heating device arranged in the process chamber and including: a heating body configured to carry a wafer, a heating member configured to generate heat and being arranged in the heating body; and a cooling structure arranged below the heating body and configured to selectively perform heat exchange with the

heating body at different positions away from the heating body, and including: at least one cooling body, a cooling member configured to perform heat exchange with the heating body being arranged in the at least one cooling body; and a cooling ascending and descending assembly connected to the at least one cooling body and configured to drive the at least one cooling body to ascend and descend to adjust a distance between the at least one cooling body and the heating body in a vertical direction; a connection member connected to the heating body, the connection member and the heating body forming accommodation space at a bottom of the heating body, the at least one cooling body being located in the accommodation space, and the cooling ascending and descending assembly being extended to outside of the process chamber passing through a through-hole arranged at a bottom of the process chamber; and a sealing structure, an upper end of the sealing structure being sealed with and connected to the connection member, and a lower end of the sealing structure being sealed with and connected to the bottom of the process chamber to seal the accommodation space and the through-hole.

12. The semiconductor apparatus according to claim 1, wherein the sealing structure includes: a bellows shaft, an upper end of the bellows shaft being connected to the connection member, and a lower end of the bellows shaft being extended to the outside of the process chamber passing through the through-hole to be connected to an ascending and descending drive source; and a bellows sleeved at the bellows shaft, an upper end of the bellows being sealed with and connected to the connection member, and a lower end of the bellows being sealed with and connected to the bottom of the process chamber to seal the accommodation space and the through-hole.

13. The semiconductor apparatus according to claim 12, wherein the cooling ascending and descending assembly further includes: a lift member located in hollow space, the hollow space communicating with the accommodation space being arranged in the bellows shaft, and an upper end of the lift member being connected to the cooling body; a screw, a threaded hole being arranged at a bottom surface of the lift member, at least one mounting hole being arranged at an inner surface of the bellows shaft opposite to the bottom surface of the lift member, and the mounting hole being coaxially arranged with the threaded hole in a one-to-one correspondence, the screw being threadedly connected to the corresponding threaded hole passing through the mounting hole from the bottom of the bellows shaft to top in a one-to-one correspondence; and an elastic member arranged in a one-to-one correspondence with the screw, located between the bottom surface of the lift member and the inner surface of the bellows shaft, and being in a compressed state.

14. The semiconductor apparatus according to claim 13, wherein: lead-out space and two first lead-out through-holes communicating with the lead-out space are arranged in the lift member; two second lead-out through-holes communicating with the accommodation space are arranged in the bellows shaft, and the two second lead-out through-holes are coaxially arranged with the two first lead-out through-holes; the cooling member further includes: a cooling pipeline configured to transfer cooling water; a water inlet pipeline; and a water outlet pipeline, an end of the water inlet pipeline and an end of the water outlet pipeline being connected to an inlet and an outlet of the cooling pipeline, respectively, and the other end of the water inlet pipeline and the other end of the water outlet pipeline being extended to the outside of the process chamber passing through the corresponding first lead-out through-hole and the second lead-out through-hole in sequence.

15. The semiconductor apparatus according to claim 11, wherein the at least one cooling body includes: a plurality of cooling bodies arranged at intervals in a direction away from a bottom surface of the heating body; and the cooling member configured to perform heat exchange with the heating body and being arranged in each cooling body of the plurality of cooling bodies.

16. The semiconductor apparatus according to claim 15, wherein: the plurality of cooling bodies are located in the accommodation space, the cooling member including: a cooling pipeline configured to transfer cooling water, a water inlet pipeline; and a water outlet pipeline, an end of the water inlet pipeline and an end of the water outlet pipeline are connected to an inlet and an

outlet of the cooling pipeline, respectively, and the other end of the water inlet pipeline and the other end of the water outlet pipeline are extended to the outside of the process chamber passing through a through-hole arranged at the bottom of the process chamber.
